



**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani**  
**Pilani Campus**  
**AUGS/ AGSR Division**

**FIRST SEMESTER 2026-27**  
**COURSE HANDOUT (Part II)**

**Date: 01/08/2026**

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

**Course No.** : PHA G623  
**Course Title** : Pharmaceutical Applications of Polymers and Biopolymers  
**Instructor-in-Charge** : Aniruddha Roy

**1. Scope and Objective of the Course:**

This course includes the analysis of polymers and biopolymers at a molecular scale, and design of devices used in contact with biological systems. Topics include molecular interactions between bio- and synthetic molecules and surfaces; design, synthesis, and processing approaches for materials that control cell functions; and application of state-of-the-art materials science to problems in tissue engineering, drug delivery, vaccines, and cell-guiding surfaces. This course covers the fundamental concepts (physical chemistry, chemistry, and materials physics, and biology) behind the design at a molecular scale of synthetic materials with biological functions.

**2. Text Book :**

**4. Course Plan**

| Lec. No. | Topics                                                                 | Contents                                                                                                                                                                                                                                                                 | Ref. |
|----------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1-3      | Overview of Polymeric Biomaterials                                     | Introduction to biomaterials science as multidisciplinary approach; Orientation to current approaches and types of molecularly designed biomaterials; Basic principles of biopolymers; Tailoring biopolymer structure and composition for biomedical application.        |      |
| 4-6      | Properties of Biomaterials                                             | Classes of polymers used in medicine; The Nature of Matter and Materials; Bulk and Surface Properties and Surface Characterization of Biomaterials; Role of water in biomaterials;                                                                                       |      |
| 7-10     | Classes of Materials Used in Medicine                                  | The Materials Side of the Biomaterials Relationship; Polymers: Basic Principles; The Organic Matrix of Restorative Composites and Adhesives; Applications of "Smart Polymers" as Biomaterials                                                                            |      |
| 11-20    | Key Concepts in the Use of Biomaterials in Surgery and Medical Devices | Adsorbed Proteins on Biomaterials; Cells and Surfaces in Vitro; Functional Tissue Architecture, Homeostasis, and Responses to Injury; The Extracellular Matrix and Cell–Biomaterial Interactions; Effects of Mechanical Forces on Cells and Tissues;                     |      |
| 21-25    | Host Reaction to Biomaterials and Their Evaluation                     | Introduction to Biological Responses to Materials; Inflammation, Wound Healing, Foreign-Body Response, and Alternative Tissue Responses; The Immune Response to Foreign Materials; Tumorigenesis and Biomaterials; Biofilms, Biomaterials, and Device-Related Infections |      |
| 26-28    | Characterization of Biomaterials                                       | Introduction to Testing Biomaterials; Concept and Assessment of Biocompatibility; In Vitro and In-vivo Assessment of Cell and Tissue Compatibility;                                                                                                                      |      |
| 29-32    | Engineering Biological Recognition of Biomaterials                     | Biological Recognition in Vivo; Protein-resistant and Cell-resistant Surfaces; Engineering Recognition of Biomaterials: Adhesion and                                                                                                                                     |      |



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|       |                                                               |                                                                                                                                                                                                                                                                                                                |  |
|-------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|       |                                                               | Migration; Cytokine Signaling from Biomaterials, Mimicking Cell-cell Contacts with Surfaces                                                                                                                                                                                                                    |  |
| 33-35 | Introduction to Applications of Biomaterials                  | Cardiovascular Medical Devices; Artificial Organs; Burn Dressings and Skin Substitutes; Biomaterials for Immunoengineering; Biomaterials Design For Vaccines; Biomaterials-Based Model Systems to Study Tumor–Microenvironment Interactions; Responsive Polymers in the Fabrication of Enzyme-Based Biosensors |  |
| 36-40 | Applications of Biomaterials in Functional Tissue Engineering | Overview of Tissue Engineering Concepts and Applications; Micromechanical Design Criteria for Tissue-Engineering Biomaterials;                                                                                                                                                                                 |  |

**5. Evaluation:**

| Component                 | Duration | Weightage (%) | Date & Time     | Remarks |
|---------------------------|----------|---------------|-----------------|---------|
| Mid-Semester Test         | 90 Min.  | 30            | <TEST_1>        | CB      |
| Comprehensive Examination | 3 h      | 40            | <TEST_C>        | CB + OB |
| *Continuous Assessment    |          | 30            | During semester |         |

\*Continuous assessment will be based on theory covered in class. Topics and number will be announced in the class. It will be in terms of home assignments, tutorials, and surprise tests.

**6. Grading Procedure:** Grading would be done by the bunching procedure. In borderline cases subjective judgment will be used to award the grades. It is not mandatory to award all the eight grades (i.e. from A to E). Subjective judgment based on attendance for the lectures, tutorials, appearance in quiz, student's involvement in the course and performance in the class would be used in the award of grades. **The student shall not be considered as "exposed" to the course, unless he/she demonstrates appreciable skill in both the class and theory components of the course and through classroom participation.** Attending tutorial and appearing surprise quiz is very important. Students not appeared in any quiz or not submitted assignments, not appeared in any test/ comprehensive exam will not be considered exposed to the course. It is also expected that student will attend classes regularly to get proper exposure and to provide chance of evaluation of his knowledge.